

the death rate must increase, or both. Later in the chapter, we'll consider some of the factors affecting these rates, including the presence of disease, predation, and limited amounts of food and other resources.

The Logistic Model and Real Populations

The growth of laboratory populations of some small animals, such as beetles and crustaceans, and of some microorganisms, such as bacteria, *Paramecium*, and yeasts, fits an S-shaped curve fairly well under conditions of limited resources (**Figure 53.10a**). These populations are grown in a constant environment lacking predators and competing species that may reduce growth of the populations, conditions that rarely occur in nature.

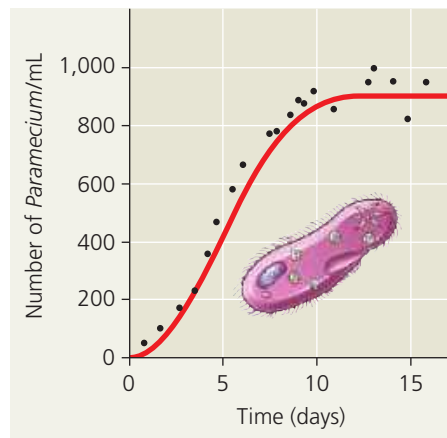
Populations in nature rarely match the predictions of the logistic model as closely as do some laboratory populations, such as the *Paramecium* population in Figure 53.10a. This is not surprising, since some of the assumptions built into the logistic model clearly do not apply to all populations. The logistic model assumes that populations adjust instantaneously to growth and approach carrying capacity smoothly. In reality, there is often a delay before the negative effects of an increasing population are realized. If food becomes limiting for a population, for instance, reproduction will decline eventually, but females may use their energy reserves to continue reproducing for a short time. This may cause the population to overshoot its carrying capacity temporarily, as shown for the water fleas in **Figure 53.10b**. In the **Scientific Skills Exercise**, you can model what can happen to such a population when N becomes greater than K . Other populations fluctuate greatly, making it difficult even to define carrying capacity. We will examine some possible reasons for such fluctuations later in the chapter.

The logistic model provides a useful starting point for thinking about how populations grow and for constructing more complex models. As such, its role is similar to that played by the Hardy-Weinberg equation for thinking about the evolution of populations. The logistic model is also important in conservation biology for predicting how rapidly a particular population might increase in numbers after it has been reduced to a small size and for estimating sustainable harvest rates for wildlife populations. Conservation biologists can also use the model to estimate the critical size below which populations of certain organisms, such as the northern subspecies of the white rhinoceros (*Ceratotherium simum*), may become extinct (**Figure 53.11**).

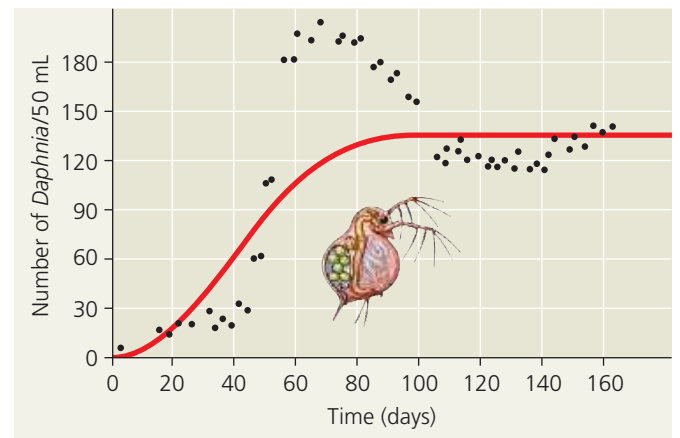
▼ **Figure 53.11 White rhinoceros mother and calf.** The two animals pictured here are members of the southern subspecies, which has a population of more than 20,000 individuals. The northern subspecies is critically endangered, with just a few known individuals.



► **Figure 53.10 How well do these populations fit the logistic growth model?** In each graph, the black dots plot the measured growth of the population, and the red curve is the growth predicted by the logistic model.



(a) A *Paramecium* population in the lab. The growth of *Paramecium aurelia* in a small culture closely approximates logistic growth if the researcher maintains a constant environment.



(b) A *Daphnia* population in the lab. The growth of a population of water fleas (*Daphnia*) in a small laboratory culture does not correspond well to the logistic model. This population overshoots the carrying capacity of its artificial environment before it settles down to an approximately stable population size.